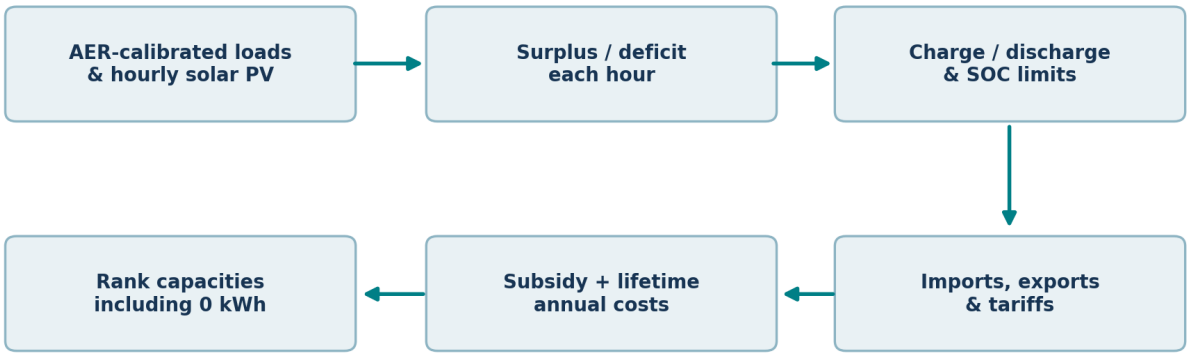
TECHNICAL RESEARCH REPORT

# Mathematical Model for Residential PV–Battery Sizing

Hourly techno-economic optimisation with verified energy accounting, power limits, SOC constraints and Australian incentive treatment




*Proposed modelling process: from household/PV energy balance to capacity selection.*

| DOCUMENT BASIS | REPORT STATUS |
|---|---|
| Input thesis | Thesis Report A new (1)(1).docx (user-supplied project draft) |
| Draft reviewed | Battery Sizing Formula Flowchart Updated (9 pages) |
| Report type | Corrected mathematical specification and illustrative verification |
| Prepared | 4 October 2026 |
| Evidence warning | Original 8,760-hour CSV data files were not supplied. Illustrative plots are NOT measured thesis battery results. |

**RESEARCH INTEGRITY** The proposed model and illustrative reconstruction must not be described as an experimentally validated optimum. Reported thesis aggregates are identified separately from illustrative figures throughout.

# Executive summary

The objective is to refine the hourly battery-sizing procedure described in the uploaded thesis and the follow-up formula flowchart. The thesis develops Australian residential load profiles and an hourly PV-generation method but primarily reports the no-battery energy mismatch. Its Section 6.17 reports annual solar generation of 10,539 kWh, household demand of 5,780 kWh, direct solar use of 2,668 kWh, PV surplus of 7,871 kWh and grid imports of 3,112 kWh [1]. These aggregate results are internally energy-consistent, but do not uniquely determine an optimal battery capacity.

The corrected formulation uses separate electrical energy and state-of-charge variables, usable-energy limits, maximum charge/discharge power, energy conservation at the AC bus, a zero-kWh baseline, cyclic-state handling, explicit tariffs and a discounted ownership cost. The principal objective selects the candidate capacity minimising annual equivalent total cost or, equivalently under the same assumptions, maximising net benefit against the PV-only baseline. A policy-aware subsidy function replaces an automatic rebate, and tariff-aware dispatch is distinguished from the baseline greedy strategy [2], [4]–[9].

A controlled, reproducible 8,760-hour illustrative reconstruction is provided only to check model behaviour and demonstrate the graphs. It is calibrated to the FIVE annual aggregate values in the uploaded thesis; this calibration does not recreate the unavailable measured/synthetic time sequence. The illustrative economic assumptions, not thesis-supplied prices, produce a minimum at 0 kWh, a valid result illustrating the possibility that battery ownership is uneconomic under a particular scenario.

| MAIN REVISION | PRACTICAL CONSEQUENCE |
|---|---|
| Power-constrained charging and discharging | Prevents impossible kWh transfer within one hour |
| Cyclic SOC and zero-battery handling | Prevents free initial energy and division by zero |
| Net-benefit vs. no-battery baseline | Explicitly values avoided imports less lost FiT |
| Discounted cost and separate subsidy eligibility | Reflects financial lifetime and evolving policy rules |
| Independent consistency tests | Provides reproducible verification criteria |

# 1. Research basis, provenance and scope

## 1.1 Findings directly supported by the thesis

The study models a 6.6 kW rooftop PV system using hourly solar irradiance (8,760 observations) with an assumed performance ratio of 0.80. It derives occupancy-based small, medium and large demand profiles with seasonal, weekday/weekend and stochastic features. The reported annual demand/solar comparison in Section 6.17 is a no-battery result for one modelled household profile [1].

| Observed or reported quantity | Value | Status |
|---|---|---|
| PV rated capacity | 6.6 kW | Thesis assumption |
| Annual irradiance | 1,996 kWh/m²/year | Thesis model output |
| PV generation | 10,539 kWh/year | Thesis result |
| Household load | 5,780 kWh/year | Thesis result |
| Direct PV to load | 2,668 kWh/year | Thesis result |
| Surplus PV | 7,871 kWh/year | Thesis result |
| Grid import, no battery | 3,112 kWh/year | Thesis result |

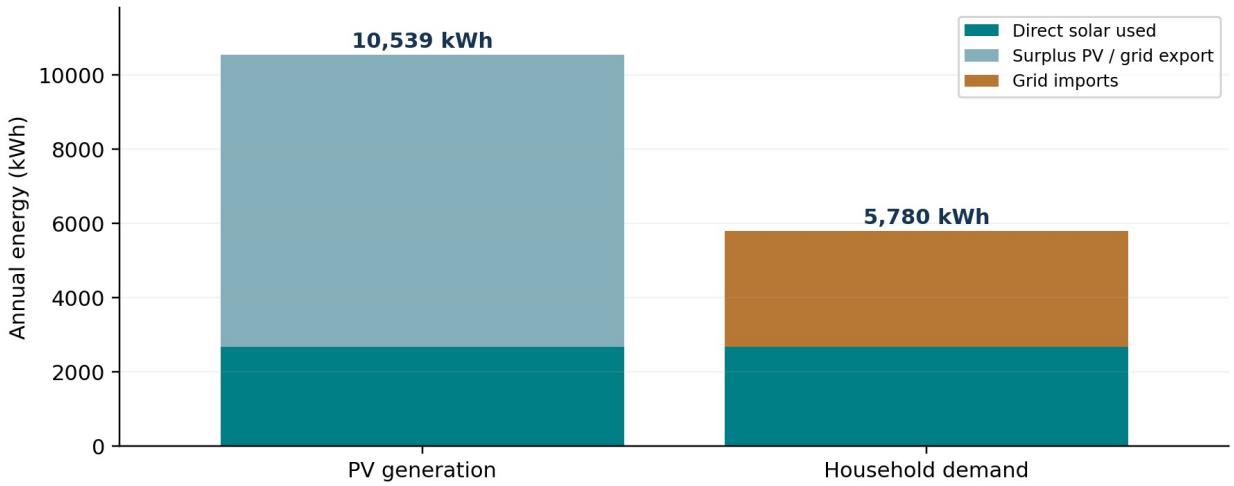



*Fig. 1. Thesis-reported no-battery annual energy partition (original results, not reconstructed).*

Annual conservation checks: 10,539 = 2,668 + 7,871 and 5,780 = 2,668 + 3,112 kWh. These identities check the aggregate accounting, not the hourly optimisation. Importantly, 46.2% is solar coverage of demand (2,668/5,780), whereas the PV self-consumption fraction is only 25.3% (2,668/10,539) under the battery-free model [1], [6].

### 1.2 Boundaries of this report

The model concerns an AC-equivalent grid-connected household PV–battery system with PV-only battery charging. Load and PV are interpreted as energy delivered at a common AC boundary after appropriate inverter/performance losses. Hardware thermal response, short-duration inrush, backup/islanding, degradation, demand charges, PV curtailment under negative FiT, advanced arbitrage and network export restrictions are not claimed as already validated by the thesis [1]. Extensions are discussed where relevant.

## 2. Technical audit of the draft

The nine-page updated flowchart correctly separates stored energy B(t) in kWh from SOC in percent and improves the battery efficiency accounting in the thesis, but additional conditions are needed before its capacity decision can be considered physically and financially complete [2].

| ISSUE | WHY IT MATTERS | CORRECTION |
|---|---|---|
| SOC initial/final condition | Unspecified starting/ending battery energy biases annual savings | Cycle annual series to a stable initial/terminal state; define B0. |
| Power rating | kWh-only constraints allow unrealistic one-hour transfers | Constrain each transfer by Pmax × Δt. |
| 0 kWh case | SOC=B/Eb becomes undefined | Bypass storage update and set all storage flows to zero. |
| PV output basis | GHI × constant PR approximates tilted PV | Record AC boundary and validate yield against radiation assumptions. |
| Grid balance/curtailment | Grid import/export rules assume nonnegative tariffs | Add optional curtailment; verify full AC bus balance. |
| Annualised capital cost | Cost/life ignores discounting, O&M, replacements | Use CRF for a simple life or full NPV over multiple years. |
| Tariff-aware dispatch | Greedy immediate discharge not globally optimal under TOU | Label rule-based benchmark; solve scheduling separately for TOU. |
| Subsidy eligibility | Draft implies subsidy for all E_b | Use function of usable capacity, installed date and policy criteria. |
| Selection set | 20 kWh extends initial 0–15 kWh thesis range | State extension explicitly and refine candidate grid. |
| Metrics in thesis | Self-consumption and payback wording inverted | Repair denominator definitions and payback orientation. |

# 3. Notation and baseline assumptions

| Symbol / term | Definition | Units |
|---|---|---|
| t, Δt | Hourly index t = 1 … 8760; interval duration | h |
| P_PV, g_t | Rated PV capacity, specific hourly PV yield | kW, kWh/kW |
| E_PV,t, E_L,t | AC-equivalent PV generation and household consumption | kWh |
| S_t, D_t | Pre-storage surplus and residual load deficit | kWh |
| E_b | Candidate nominal battery capacity | kWh |
| B_t | Stored energy at end of time step | kWh |
| α_min, α_max | SOC bounds on fractional scale (0–1) | dimensionless |
| η_c, η_d | Charging and discharging efficiencies | dimensionless |
| P_c,max, P_d,max | Charge/discharge power ratings | kW |
| E_c,t, E_d,t | Battery charging input and discharged AC energy | kWh |
| E_i,t, E_x,t | Grid import and export | kWh |
| T_i,t, T_FIT,t | Retail import / export credit tariff | AUD/kWh |
| C_0(E_b), S_gov(E_b) | Installed cost and applicable government discount | AUD |
| r, L | Annual discount rate, economic lifetime | per year, years |

**SIGN CONVENTION** All E quantities represent energy (kWh) within one time step. Charging energy is measured before charging loss and discharging energy is delivered to the AC load after discharging loss. SOC equations use B/E_b and are never applied to the 0-kWh baseline.

# 4. Complete proposed mathematical function

## 4.1 PV generation and temporal mismatch

$$E_{PV,t} = P_{PV} \times g_t ; \quad g_t = (G_{POA,t} / G_{STC}) \times PR_t \times \Delta t \quad (1)$$

The thesis retains 6.6 kW and PR = 0.80 for its baseline. G_POA refers to plane-of-array irradiance, whereas using the available global horizontal irradiance (GHI) without transposition is a documented approximation. Actual PV output should also respect AC inverter export/generation limits when independently known [1], [10].

$$S_t = \max(0, E_{PV,t} - E_{L,t}) ; \quad D_t = \max(0, E_{L,t} - E_{PV,t}) \quad (2)$$

For each hour, PV directly supplies the contemporaneous load before the battery acts. Thus only S_t can charge storage under PV-only charging, and only D_t can be replaced by battery discharge. Positive S_t and D_t cannot occur simultaneously.

## 4.2 Corrected battery limits and state transition

$$E_{c,t} = \min\{ S_t, P_{c,max} \Delta t, (\alpha_{max} E_b - B_{(t-1)}) / \eta_c \} \quad (3)$$

$$E_{d,t} = \min\{ D_t, P_{d,max} \Delta t, \eta_d [B_{(t-1)} - \alpha_{min} E_b] \} \quad (4)$$

$$B_t = B_{(t-1)} + \eta_c E_{c,t} - E_{d,t} / \eta_d \quad (5)$$

$$\alpha_{min} E_b \le B_t \le \alpha_{max} E_b ; \quad SOC_t(\%) = 100 B_t / E_b \ (E_b > 0) \quad (6)$$

For the illustrated numerical experiments only, α_min = 0.10, α_max = 0.95, η_c = η_d = 0.95 and a 0.5 C-rate are assumed, i.e., P_c,max = P_d,max = 0.5E_b kW. These are transparent study assumptions, not claims that all marketed batteries have identical power ratings. Hardware-specific AC power ratings, inverter constraints and efficiencies should replace them when known [9].

At E_b = 0, define E_c,t = E_d,t = B_t = 0 without calculating SOC. For E_b > 0, simulate the 8,760-hour series to a repeatable cyclic state such that B_8760 ≈ B_0, or impose a suitable terminal energy constraint. This avoids giving an artificial benefit to an initially full battery.

### 4.3 Grid exchange and energy conservation

$$E_{i,t} = D_t - E_{d,t}\ ; \quad E_{x,t} = S_t - E_{c,t} - E_{curt,t} \quad (7)$$

$$E_{PV,t} + E_{i,t} + E_{d,t} = E_{L,t} + E_{c,t} + E_{x,t} + E_{curt,t} \quad (8)$$

In the nonnegative-FiT baseline, curtailment E_curt,t is fixed at zero. If exporting is disallowed, capped or charged a negative price, a curtailment control variable may be enabled, with $0 \le E_{curt,t} \le S_t - E_{c,t}$ and separate export-cap constraints. Equation (8) must equal zero residual in every simulated hour.

### 4.4 Annual grid cost and ownership cost

$$C_{grid}(E_b) = \Sigma_{(t=1\ldots8760)}[\ T_{i,t}\, E_{i,t} - T_{FIT,t}\, E_{x,t}\ ] \quad (9)$$

$$C_{net}(E_b) = C_0(E_b) - S_{gov}(E_b) \quad (10)$$

$$CRF(r,L) = r(1+r)^L / [(1+r)^L - 1]\ ; \quad CRF(0,L) = 1/L \quad (11)$$

$$AEC(E_b) = CRF(r,L)\, C_{net}(E_b) + C_{O\&M}(E_b) \quad (12)$$

AEC is an equivalent annual ownership cost. The gross installed cost must include the whole installed battery and any incremental inverter/electrical work, not just cell cost. Fixed household supply charges cancel in battery-versus-no-battery comparisons only when tariff schedules do not change with capacity. Full NPV with replacements, salvage and degradation is recommended for long-term investment analyses.

### 4.5 Optimisation objective and equivalent benefit form

$$E_b^* = \arg\min_{E_b \in \mathcal{B}} \{\ C_{grid}(E_b) + AEC(E_b)\ \} \quad (13)$$

The candidate set $\mathcal{B}$ must include 0 kWh and physically marketable storage capacities. $\mathcal{B}$ can initially be {0, 5, 10, 13.5, 15} kWh to match the thesis and later be refined, but results can only be called the best of the tested capacities unless a sufficiently broad/continuous optimisation has been demonstrated.

$$NB(E_b) = C_{grid}(0) - [C_{grid}(E_b) + AEC(E_b)] \quad (14)$$

$$NB(E_b) = \Sigma[T_{i,t}\, E_{d,t} - T_{FIT,t}\, E_{c,t}] - AEC(E_b) \quad (15)$$

Equation (15) follows from Eq. (7) when curtailment is zero and there is no grid charging. Therefore $E_b^* = \arg\max NB(E_b)$. The difference between baseline grid imports and battery-case imports equals discharge; lost PV exports equal charging. When grid charging or curtailment is allowed, Eq. (14) remains general, but Eq. (15) needs additional terms. A battery may be economically rejected when 0 kWh yields the largest net benefit [4], [5].

**FINAL RECOMMENDED FUNCTION** For the present thesis scope, present Eqs. (1)–(15) as the proposed hourly simulation-based sizing model, with Eq. (13) the formal cost-minimisation decision and Eq. (15) the interpretable net-benefit equivalent under specified dispatch assumptions.

## 5. Tariff dispatch and Australian incentives

### 5.1 Why a price-aware extension matters

The charge/discharge min rules in Eqs. (3) and (4) intentionally reproduce the thesis priority logic: direct PV use, excess PV to battery, residual PV to grid, then deficit supplied by battery and grid. This greedy rule is suitable as a transparent baseline, but is not necessarily optimal under varying import tariffs because

energy might be more valuable at a later evening peak. It should not be described as globally optimal dispatch merely because the outer loop chooses the cheapest capacity [5], [9], [11].

A TOU-aware extension treats $E_{c,t}$, $E_{d,t}$, $B_t$, $E_{i,t}$ and $E_{x,t}$ as decision variables and, for each $E_b$, minimises $\Sigma[T_{i,t} E_{i,t} - T_{FIT,t} E_{x,t}]$ over the horizon, subject to the SOC, power, energy balance and permitted grid/PV source restrictions. Enforce $E_{c,t} \leq S_t$ when PV-only charging is intended, and impose non-simultaneous charging/discharging where needed. The economic sizing model then compares optimally dispatched capacities with the same financial objective. TOU results must be explicitly separated from the original rule-based thesis baseline.

## 5.2 Modelling the 2026 subsidy without invented amounts

The Australian Government Cheaper Home Batteries Program now provides a discount for eligible small-scale batteries. From 1 May 2026, the incentive is based on eligible usable battery capacity, the installation-date STC factor and tiered capacity treatment; the posted factor for May–December 2026 is 6.8, with 100% of the factor on the first 14 kWh of usable capacity, 60% for the next 14 kWh and 15% above 28 to 50 kWh [7], [8]. Eligible batteries have additional nominal-capacity, product, installer and other requirements. Eligibility cannot be assumed from nominal $E_b$ alone.

$$u(E_u) = \min(E_u,14) + 0.60 \cdot \min(\max(E_u - 14,0),14) + 0.15 \cdot \min(\max(E_u - 28,0),22) \quad (16)$$

For a hypothetical eligible installation in May–December 2026, an indicative certificate-weighted expression is $STC_{raw} = 6.8 \times u(E_u)$, where $E_u$ is the policy-defined eligible usable capacity. Actual certificates, rounding, pricing, deeming and eligibility must be determined using the current regulator/REC Registry calculator and installer quote. Do NOT treat $STC_{raw}$ × an invented dollar value as a verified rebate [7], [8].

$$S_{gov}(E_b) = \{ \text{verified quote discount}(E_u,\text{date},\text{eligibility}), \text{if eligible; } 0 \text{ otherwise} \} \quad (17)$$

This formulation allows the current 2026 program to be incorporated without silently equating 30% of the nominal battery price with the legislated calculation. The simulation figures below use a separate, plainly hypothetical flat 30% cost reduction solely to illustrate a financial sensitivity, not an Australian policy quote.

# 6. Validation protocol and worked example

## 6.1 Hourly implementation algorithm

| STEP | PROCEDURE | EXPECTED ACTION |
|---|---|---|
| 1 | Set scenario inputs | Select household load and 6.6 kW PV profile; define all financial and equipment assumptions. |
| 2 | Run 0-kWh baseline | Compute direct use, PV export, grid import and baseline grid bill without SOC. |
| 3 | Choose candidate $E_b$ | Set battery SOC bounds, efficiencies, charge/discharge power limits and initial condition. |
| 4 | Find cyclic SOC | Repeat the identical annual sequence or solve a cycle-boundary condition to eliminate free starting energy. |
| 5 | Simulate each hour | Compute PV surplus/deficit, enforce charge/discharge constraints, update $B_t$. |
| 6 | Check equations | Track physical energy-balance residual, SOC limits and Pmax constraints. |
| 7 | Compute economics | Evaluate grid bill, verified subsidy when applicable, discounted annual cost and net benefit. |
| 8 | Select and test | Rank sizes, refine grid near minimum and stress-test tariffs, occupancy, solar year and ageing. |

## 6.2 Single-hour illustrative arithmetic

Consider a hypothetical 10 kWh battery with initial stored energy $B(t-1) = 4.00$ kWh, $\eta_c = \eta_d = 0.95$, SOC bounds 10–95% and Pmax = 5 kW. In the first hour, PV = 5.0 kWh and load = 1.5 kWh, yielding $S_t = 3.5$ kWh. By Eq. (3), $E_{c,t} = \min(3.5, 5, (9.5 - 4)/0.95) = 3.5$ kWh. Thus $B_t = 4 + 0.95(3.5) = 7.325$ kWh and SOC = 73.25%; export = 0. In the next hour, with PV = 0 and load = 4.0 kWh, discharge = $\min(4, 5, (7.325 - 1)\times 0.95) = 4.0$ kWh. $B_t = 7.325 - 4/0.95 = 3.114$ kWh, SOC ≈ 31.14%, and import = 0. All figures are hypothetical checks of the equations, not thesis observations.

## 6.3 Verification rules

| CHECK | TEST | RESULT IN ILLUSTRATION |
|---|---|---|
| Annual no-battery balance | PV = direct PV + exports; load = direct PV + imports | PASS: 10,539 and 5,780 kWh |
| Every-hour bus balance | PV + import + discharge = load + charge + export + curtail | PASS: residual below $10^{-8}$ kWh |
| SOC safety | $\alpha_{min} E_b \le B_t \le \alpha_{max} E_b$ | PASS: all tested capacities |
| Power rating | $E_{c,t}, E_{d,t} \le$ Pmax × 1 h | PASS: all tested capacities |
| Zero PV stress | No sun should not generate battery energy | PASS |
| Zero-capacity baseline | No division by zero; no battery flows | PASS |
| Cyclic-boundary condition | No unaccounted opening/closing battery energy | PASS within $10^{-6}$ kWh |
| Objective equivalence | Baseline difference = avoided imports − lost FiT − ownership | PASS < $10^{-7}$ AUD |

The checks were executed against the illustrative reconstructed input series described next. Passing these checks validates internal implementation consistency only. It does not validate the empirical accuracy of missing original hourly profiles or market quotes.

# 7. Illustrative numerical experiment and graphs

## 7.1 Evidence separation and construction

The original 8,760-hour source arrays are not attached to the thesis file. Therefore a deterministic, illustrative 8,760-hour PV and demand sequence was constructed with Australian-like summer/winter PV variation, morning/evening demand peaks, weekend variation and fixed random seed 42. The proxy was rescaled and its daytime load shape adjusted to exactly reproduce the thesis annual PV generation (10,539), load (5,780), directly consumed PV (2,668), surplus (7,871) and grid import (3,112) kWh. Matching these aggregate constraints DOES NOT mean the reconstructed sequence is the thesis sequence; an unbounded number of differently timed sequences have identical totals but different battery optima.

For illustration only, a flat retail tariff of AUD 0.35/kWh, FiT of AUD 0.05/kWh, discount rate 6%, economic life 10 years, installed cost of AUD 1,000 fixed plus AUD 700/kWh, annual O&M of 1% of gross installed cost and hypothetical 30% capital discount are assumed. PV-only charging, $\eta_c = \eta_d = 0.95$, SOC 10–95%, 0.5 C-rate and annual cyclic-state enforcement are applied. NONE of these illustrative financial values is represented as a measured thesis result, retailer quote or verified program rebate.

| ILLUSTRATIVE INPUT | VALUE | EVIDENCE CLASS |
|---|---|---|
| 8760 hours; annual PV/load/overlap | 10,539 / 5,780 / 2,668 kWh | Aggregate-calibrated to thesis |
| Charge/discharge efficiency | 0.95 / 0.95 | Assumption |
| SOC / power | 10–95% / 0.5 C-rate | Assumption |
| Tariff / feed-in credit | AUD 0.35 / 0.05 per kWh | Hypothetical tariff |
| Gross battery capex | AUD 1,000 + AUD 700/kWh | Hypothetical cost |
| Subsidy for demonstration | 30% of gross capex | Hypothetical, not actual STC |
| Interest / lifetime / annual O&M | 6% / 10 years / 1% gross | Assumption |

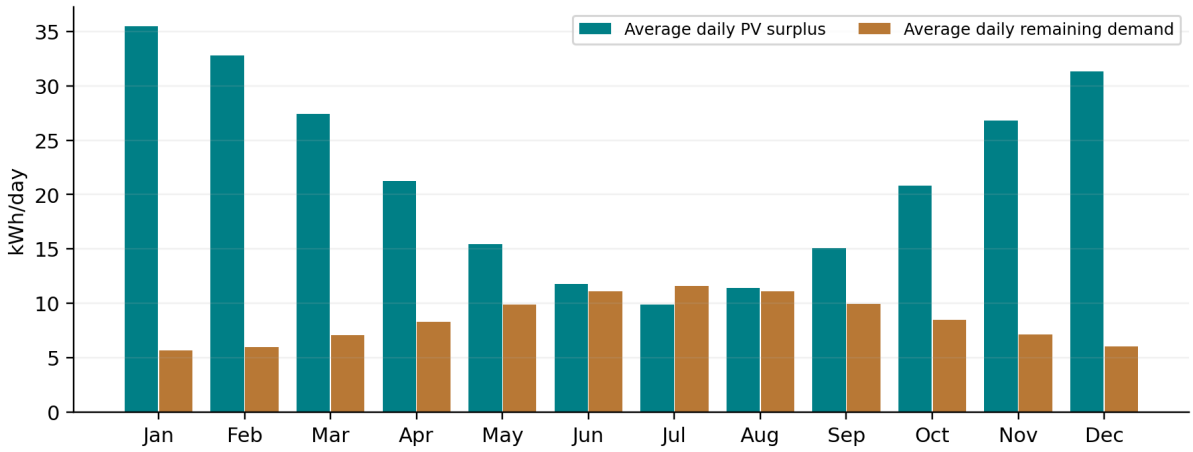



*Fig. 2. Daily surplus and deficit by month in the reconstructed 8,760-hour illustration; NOT original thesis hourly results.*

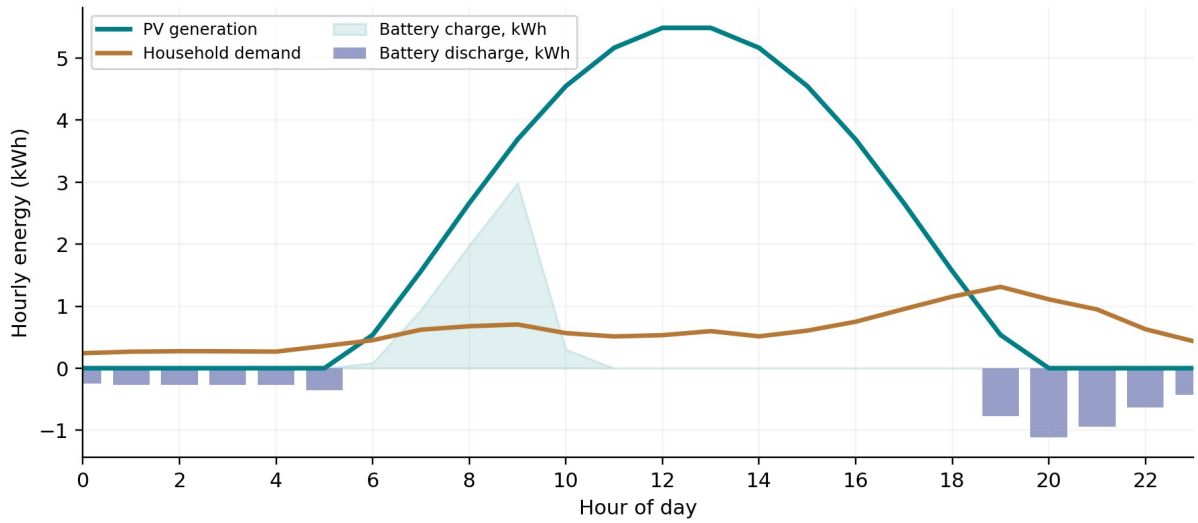



*Fig. 3. One illustrative summer day: PV, load, charge/discharge using a 10 kWh battery; NOT measured data.*

## 7.2 Capacity optimisation under example costs

| Capacity (kWh) | Grid imports (kWh) | Grid exports (kWh) | Grid cost (AUD/yr) | Total annual cost (AUD/yr) | Net benefit (AUD/yr) |
|---|---|---|---|---|---|
| 0 | 3,112 | 7,871 | 696 | 696 | +0 |
| 5 | 1,638 | 6,238 | 262 | 734 | -39 |
| 7.5 | 923 | 5,446 | 51 | 708 | -12 |
| 10 | 433 | 4,903 | -94 | 747 | -52 |
| 13.5 | 143 | 4,582 | -179 | 919 | -224 |
| 15 | 138 | 4,576 | -181 | 1,028 | -333 |
| 20 | 134 | 4,571 | -182 | 1,395 | -699 |

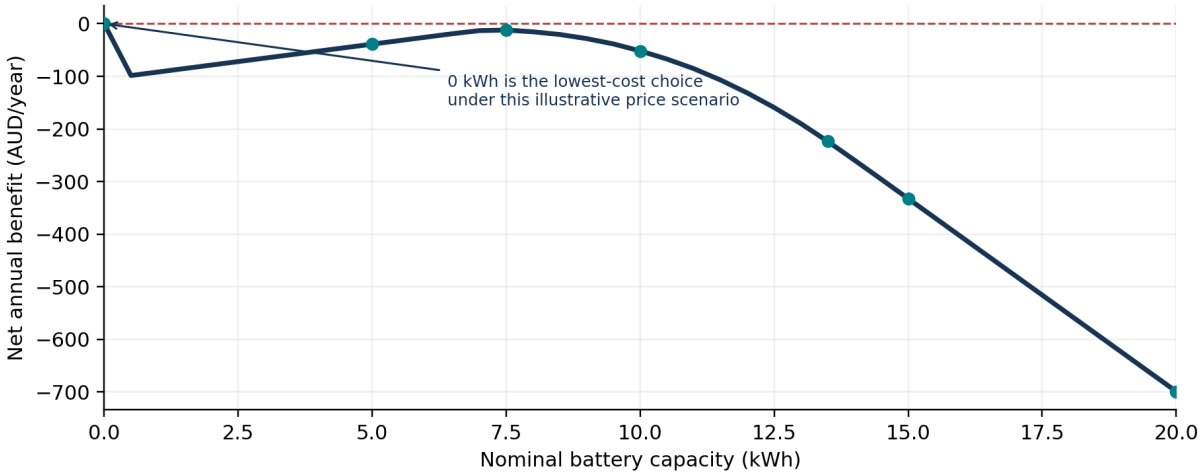



*Fig. 4. Illustrative net annual financial benefit versus battery capacity. The chosen minimum-cost point is 0 kWh under the stated prices; this is NOT the optimal size from the original thesis dataset.*

In this example, the annual PV-only grid bill is AUD 695.65. A 7.5 kWh battery saves imported electricity and increases self-consumption, yet costs around AUD 12.07/year more than the PV-only case once ownership costs are annualised. Consequently 0 kWh is selected from the illustrated candidate grid. This result is scenario-specific and must not be generalised to actual Australian households or treated as a finding of the uploaded thesis.

### 7.3 Sensitivity to tariff assumptions

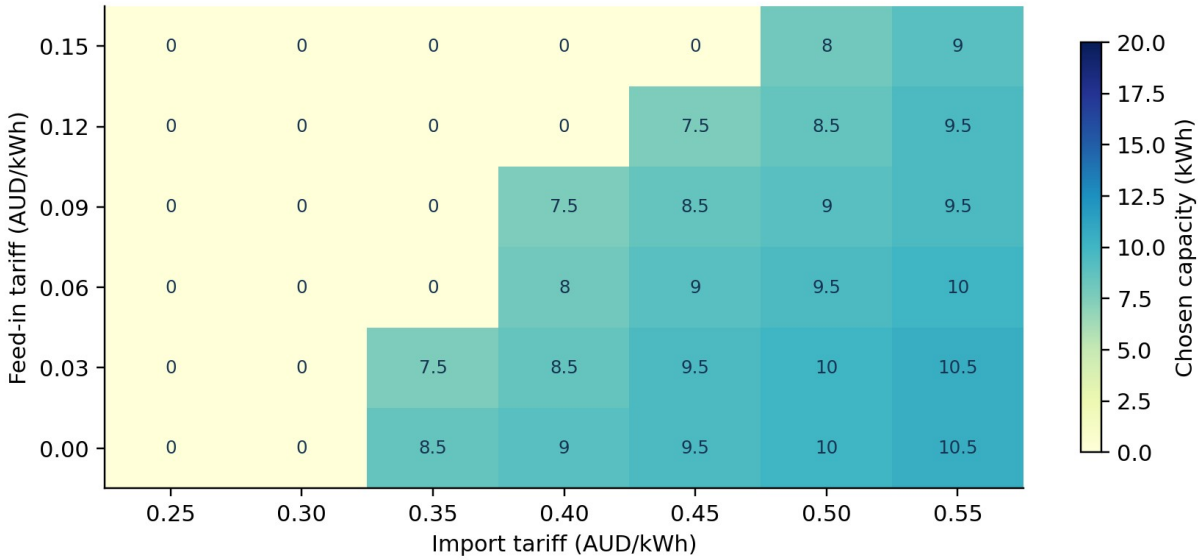



*Fig. 5. Selected battery capacity (kWh) for different hypothetical import tariffs and FiTs, holding other illustrative parameters fixed; NOT empirically estimated tariff predictions.*

The illustration demonstrates why no standalone constant capacity such as “10 kWh for all 6.6 kW systems” is justified. Even when aggregate PV and load totals are held fixed, relative import/FIT values can change the economic minimum. A positive selected size at some tariff points should be interpreted as the best capacity for the assumed proxy hourly series, not necessarily for the thesis’s original profile.

## 8. Interpretation, screening rules and robustness

### 8.1 Marginal economic value and feasibility

$$\eta_{RT} = \eta_c \eta_d ; \quad \text{storing 1 kWh PV is useful when } \eta_{RT} T_{import,later} > T_{FIT,now} \tag{18}$$

This marginal rule screens the operating value of shifting surplus PV to a later deficit in the absence of capacity/power restrictions; with 0.95 charge and discharge efficiency, $\eta_{RT} = 0.9025$. It is not a sufficient investment decision rule because installed cost, lifetime, utilisation and changing tariffs must also be included. A load deficit beyond 3,112 kWh/year avoided imports is impossible under the baseline annual demand partition and PV-only charging constraints; actual reductions are smaller owing to energy losses and timing limitations.

### 8.2 Required sensitivity experiments on the actual thesis data

| FACTOR | SUGGESTED STUDY GRID | WHY |
|---|---|---|
| Household occupancy | Small / medium / large | Tests occupancy dependency already in thesis |
| PV size | 4 / 6.6 / 8 / 10 kW | Tests PV capacity effect |
| Battery size | 0–20 kWh plus fine grid near minimum | Avoids candidate-grid artefacts |
| Charge/discharge powers | 0.25 / 0.5 / 1 C and manufacturer ratings | Avoids unreal transfer rates |
| Efficiency | 90–98% per direction, sourced ranges | Captures loss sensitivity |
| Tariffs | Actual retailer schedule + alternative TOU | Separates price and dispatch effects |
| FiT | Current applicable provider plan(s) | Captures opportunity cost |
| Installed cost & subsidy | Actual quotes and policy-verified STCs | Avoids unsupported assumptions |
| Data sensitivity | Multiple random seeds + measured solar years | Shows robustness of synthetic data results |
| Lifetime & degradation | Nominal, high/low lifetime, replacement year | Shows investment uncertainty |

A useful uncertainty summary would report the fraction of synthetic data realisations selecting each capacity, the cost penalty of rounding to commercial battery sizes, and the difference between rule-based dispatch and tariff-aware dispatch where applicable. This should be discussed as model robustness rather than experimental certainty.

## 9. Specific changes recommended for the existing thesis

- Replace the unscaled SOC recurrence in Section 2.12 and Section 3.8.7 with stored-energy Eq. (5) and percentage Eq. (6). The current prose mixes kWh with a quantity named SOC.
- Correct Section 3.8.11: 25.3% is baseline PV self-consumption, while 46.2% is the fraction of household demand directly covered by PV. Give distinct formulas and symbols.
- Correct simple payback to net initial investment divided by positive annual bill savings less incremental annual O&M. Do not invert it or double-count annualised battery capex.
- Update Section 3.8.9 with CRF/NPV, eligibility-aware STC functions and clear AUD units. If policy changes are not a formal original objective, label them an extension.
- Revise the Results and Conclusion to separate demonstrated annual solar-load mismatch from future/proposed battery optimisation. Do not describe an uncomputed optimum as demonstrated.
- Include hourly validation plots for minimum/maximum SOC, battery rated power, cyclic start/end SOC, grid imports and exports alongside financial sensitivity plots.
- Use measured/actual hourly CSV input datasets in the final experiments; record weather location/year, data provenance, installation date, tariff plan, quote date and random seed.

# Appendix A. Reproducible model pseudocode

```
FOR each E_b in candidate capacities including E_b = 0:
    IF E_b == 0:
        E_charge = E_discharge = 0; do not calculate SOC
    ELSE:
        set nominal capacity, SOC bounds, eta_c/d and kW limits
        cycle the same annual input until start/end B converge
        FOR each hour t=1...8760:
            PV = P_PV*g[t]; surplus=max(PV-load[t],0)
            deficit=max(load[t]-PV,0)
            if surplus>0: charge = min(surplus,Pch*dt,room/eta_c)
            else:          discharge = min(deficit,Pdis*dt,
                                                available*eta_d)
            B = B + eta_c*charge - discharge/eta_d
            import = deficit-discharge
            export = surplus-charge-curtailment
            assert energy balance, SOC and power limits
    C_grid = sum(import*T_import - export*T_FIT)
    C_installed = full equipment + installation cost
    C_net = C_installed - eligible_policy_discount
    annual_ownership = CRF*C_net + annual_OandM
    total_cost[E_b] = C_grid + annual_ownership
CHOOSE the E_b giving lowest total_cost
RERUN all critical physical and sensitivity checks
```

# Appendix B. Required input and output schema

| INPUT DATA COLUMN | REQUIRED TREATMENT |
|---|---|
| timestamp | Precisely 8,760 consecutive hours for a non-leap year; record time zone and DST convention |
| household_load_kwh | Non-negative AC demand each hour |
| pv_generation_kwh or ghi_w_m2 | Non-negative AC PV energy or raw radiation + orientation/performance assumptions |
| import_tariff_aud_kwh | Hourly tariff; include controlled load charges as relevant |
| feed_in_tariff_aud_kwh | Hourly or retailer-defined exported kWh credit |
| battery_nominal_kwh | Commercial model choices including 0 |
| battery_usable_kwh / power_kw | Manufacturer kWh and charge/discharge AC ratings |
| battery_cost_aud / stc_quote_aud | Quote-verified installed cost and available discount |
| assumption_metadata | Location; weather year; load scenario; seed; tariff plan; purchase date |
| **OUTPUT** | **FORMULA OR DEFINITION** |
| grid import/export | $\Sigma E_{i,t}$ and $\Sigma E_{x,t}$ (kWh/year) |
| solar self-consumption | 100 × (PV directly used + PV allocated to charging) / total PV, clearly defining loss accounting |
| solar self-sufficiency | 100 × (direct PV + discharge delivered to load) / annual load |
| equivalent full cycles | $\Sigma(E_{d,t}/\eta_d)/E_b$ for $E_b > 0$ |
| payback years | Net installed cost / positive annual gross bill savings net of O&M |
| bill savings | $C_{grid}(0) - C_{grid}(E_b)$ |
| net annual benefit | $C_{grid}(0) - C_{grid}(E_b) - AEC(E_b)$ |

## References (IEEE style)

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Source integrity note: [1] and [2] are user-provided working drafts, not peer-reviewed external publications. The reference list distinguishes them from peer-reviewed sources and public Australian Government or NREL technical materials.